

Year 2 Quantum Mechanics - Solutions to Example Sheet 2

1. Expectation values:

$$\langle x \rangle = \int_{-a}^a \psi^*(x) x \psi(x) dx = \frac{1}{a} \int_{-a}^a x \cos^2(\pi x/2a) dx = 0$$

$$\langle x^2 \rangle = \int_{-a}^a \psi^*(x) x^2 \psi(x) dx = \frac{1}{a} \int_{-a}^a x^2 \cos^2(\pi x/2a) dx = \frac{1}{a} \left[\frac{x^3}{6} + \left(\frac{ax^2}{2\pi} - \frac{a^3}{\pi^3} \right) \sin(\pi x/a) + \frac{a^2 x \cos(\pi x/a)}{\pi^2} \right]_{-a}^a = \frac{a^2}{3} - \frac{2a^2}{\pi^2}$$

$$\langle p \rangle = \int_{-a}^a \psi^*(x) \left(-i\hbar \frac{d\psi(x)}{dx} \right) \psi(x) dx = \frac{-i\hbar}{a} \int_{-a}^a \left(\frac{-\pi}{2a} \right) \cos(\pi x/2a) \sin(\pi x/2a) dx = 0$$

$$\langle p^2 \rangle = \int_{-a}^a \psi^*(x) \left(-\hbar^2 \frac{d^2\psi(x)}{dx^2} \right) \psi(x) dx = \frac{\hbar^2 \pi^2}{4a^3} \int_{-a}^a \cos^2(\pi x/2a) dx = \frac{\hbar^2 \pi^2}{4a^3} \left[\frac{x}{2} + \frac{a}{2\pi} \sin(\pi x/a) \right]_{-a}^a = \frac{\hbar^2 \pi^2}{4a^2}$$

$$\sqrt{\langle x^2 \rangle \langle p^2 \rangle} = \sqrt{\left(\frac{a^2}{3} - \frac{2a^2}{\pi^2} \right) \frac{\hbar^2 \pi^2}{4a^2}} = \sqrt{\frac{\hbar^2 \pi^2 (\pi^2 - 6)}{12\pi^2}} = 0.57\hbar$$

Interpretation: since $\langle x \rangle = 0$ and $\langle p \rangle = 0$, then $\sqrt{\langle x^2 \rangle}$ and $\sqrt{\langle p^2 \rangle}$ are equal to the RMS deviations of x and p respectively, i.e. σ_x and σ_p , and so $\sigma_x \sigma_p = 0.57\hbar > \hbar/2$

2. Velocity of proton is $0.1c$, so kinetic energy is $T = mv^2/2 = \frac{1.67 \times 10^{-27}}{2} \times (3 \times 10^7)^2 = 7.5 \times 10^{-13} J = 4.7$ MeV.

In region before barrier, potential = 0 so $E = T = 7.5 \times 10^{-13} J$.

Past step, $V = 10$ MeV = $1.6 \times 10^{-12} J$.

In excluded region, wavefunction has the form $\psi(x) = Ae^{-Kx}$ where

$$K = \sqrt{2m(V - E)}/\hbar = \sqrt{2 \times 1.67 \times 10^{-27} \times 8.5 \times 10^{-13}} / 1.05 \times 10^{-34} \text{ m}^{-1} = 5.1 \times 10^{14} \text{ m}^{-1}.$$

Probability within excluded region is then $P(x) = |\psi(x)|^2 = A^2 e^{-2Kx}$.

So the penetration depth = $1/2K = 9.9 \times 10^{-16} \text{ m}$.

3. Kinetic energy of electron is $2 \text{ eV} = 3.2 \times 10^{-19} J$.

Height of barrier = $4 \text{ eV} = 6.4 \times 10^{-19} J$.

In barrier region, wavefunction has exponential form where $K = \sqrt{2m(V - E)}/\hbar = 7.3 \times 10^9 \text{ m}^{-1}$.

The barrier transmission coefficient is

$$T = 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right) e^{-2Ka} = 16 \frac{1}{2} \left(1 - \frac{1}{2} \right) e^{-2 \times 7.3 \times 10^9 \times 10^{-9}} = 4e^{-14.6} = 2 \times 10^{-6}.$$

4. a) Normalise by requiring $\int_0^L \psi(x)^* \psi(x) dx = 1$.

$$\begin{aligned} 1 &= A^2 \int_0^L 4 \sin^2(\pi x/L) + \sin^2(2\pi x/L) + 4 \sin(\pi x/L) \sin(2\pi x/L) dx \\ &= A^2 \int_0^L 2 - 2 \cos(2\pi x/L) + \frac{1}{2} - \frac{1}{2} \cos(4\pi x/L) dx = A^2 \left[\frac{5}{2}x - \frac{L}{\pi} \sin(2\pi x/L) - \frac{L}{8\pi} \sin(4\pi x/L) \right]_0^L = \\ &= A^2 \frac{5L}{2} \text{ so } A = \sqrt{\frac{2}{5L}}. \end{aligned}$$

- b) The wavefunction consists of a superposition of 2 energy eigenstates:

$$\psi_1 = 2A \sin(\pi x/L) e^{-iE_1 t/\hbar}$$

$$\psi_2 = A \sin(2\pi x/L) e^{-iE_2 t/\hbar}$$

The integral of $|\psi_1(x)|^2$ from 0 to L is $4 \times$ that of $|\psi_2(x)|^2$ (we have already done these integrals in order to find the value of A), so the probabilities of measuring E_1 and E_2 are 80% and 20% respectively. From this we may estimate $\langle E \rangle = (4E_1 + E_2)/5 = 4\pi^2 \hbar^2 / 5mL^2$.

- c) the expectation value is

$$\begin{aligned} \langle E \rangle &= \int_0^L \psi^* \left(i\hbar \frac{\partial}{\partial t} \right) \psi dx = \\ &= \frac{2}{5L} \int_0^L \left(2 \sin(\pi x/L) e^{iE_1 t/\hbar} + \sin(2\pi x/L) e^{iE_2 t/\hbar} \right) \left(2E_1 \sin(\pi x/L) e^{iE_1 t/\hbar} + E_2 \sin(2\pi x/L) e^{iE_2 t/\hbar} \right) dx = \\ &= \frac{2}{5L} \int_0^L \left(4E_1 \sin^2(\pi x/L) + E_2 \sin^2(2\pi x/L) + (2E_1 + 2E_2) \sin(\pi x/L) \sin(2\pi x/L) \right) dx = \\ &= \frac{2}{5L} \left(4E_1 \frac{L}{2} + E_2 \frac{L}{2} \right) = 0.8E_1 + 0.2E_2. \end{aligned}$$

5. a) Classically the maximum displacement occurs when the potential energy = the total energy, thus:

$$\frac{1}{2}\hbar\omega = \frac{1}{2}Cx^2$$

$$x^2 = \frac{\hbar\omega}{C}$$

$$x^4 = \frac{\hbar^2\omega^2}{C^2}$$

and since the classical frequency $\omega = \sqrt{C/m}$, this gives

$$x^4 = \frac{\hbar^2 C}{mC^2} = \frac{\hbar^2}{mC} = a^4$$

The probability density $|\psi|^2 = e^{-x^2/a^2}$ is plotted in the Notes available on Canvas (section about Simple Harmonic Oscillator). The most significant observation is that the tails extend beyond the classical limit.

- b) Given $\psi(x) = (1 + bx^2)e^{-x^2/2a^2}$, then

$$\frac{d^2\psi}{dx^2} = \left[\left(2b - \frac{1}{a^2}\right) - x^2 \left(\frac{5b}{a^2} - \frac{1}{a^4}\right) + x^4 \frac{b}{a^4} \right] e^{-x^2/2a^2}$$

and substituting into the Schroedinger equation, assuming (as given) that $E = 5\hbar\omega/2$, we obtain:

$$\frac{-\hbar^2}{2m} \left[\left(2b - \frac{1}{a^2}\right) - x^2 \left(\frac{5b}{a^2} - \frac{1}{a^4}\right) + x^4 \frac{b}{a^4} \right] e^{-x^2/2a^2} + \frac{Cx^2}{2} (1 + bx^2) e^{-x^2/2a^2} = \frac{5\hbar\omega}{2} (1 + bx^2) e^{-x^2/2a^2}$$

$$\frac{-\hbar^2}{2m} \left[\left(2b - \frac{1}{a^2}\right) - x^2 \left(\frac{5b}{a^2} - \frac{1}{a^4}\right) + x^4 \frac{b}{a^4} \right] + \frac{Cx^2}{2} (1 + bx^2) = \frac{5\hbar\omega}{2} (1 + bx^2)$$

This can only be true for all x if the equation is true separately for each power of x . So from the constant terms we obtain:

$$\frac{-\hbar^2}{2m} \left(2b - \frac{1}{a^2}\right) = \frac{5\hbar\omega}{2} \text{ giving } b = \frac{-2m}{\hbar} \sqrt{\frac{C}{m}} = -\frac{2}{a^2}$$

Substitute this into the equations for the coefficients of x^2 and x^4 to verify that this gives a consistent solution. It does, so $\psi(x)$ is indeed a solution for $E = \frac{5\hbar\omega}{2}$.